

Grade 12 GS: Advanced Mechanical Oscillations Study Guide

Scope: Torsion, Compound, & Simple Pendulums

I. The Unified Harmonic Model

All three pendulum types discussed in Chapter 4 reduce to the same Second-Order Linear Differential Equation when oscillating with small amplitudes ($\theta_m \leq 10^\circ$).

The Master Equation

$$\theta'' + \omega_0^2 \theta = 0$$

- θ : Angular position (radians)
- θ'' : Angular acceleration ($\ddot{\theta}$)
- ω_0 : Proper angular frequency (rad/s)

Comparative Parameter Table

System	Potential Energy Type	Stiffness Term (K_{eq})	Proper Frequency (ω_0)	Period (T_0)
Torsion	EPE (Wire Twist)	C	$\sqrt{\frac{C}{I}}$	$2\pi\sqrt{\frac{I}{C}}$
Compound	GPE (Gravity)	Mga	$\sqrt{\frac{Mga}{I}}$	$2\pi\sqrt{\frac{I}{Mga}}$
Simple	GPE (Gravity)	mgL	$\sqrt{\frac{g}{L}}$	$2\pi\sqrt{\frac{L}{g}}$

Critical Note for Compound Pendulums:

- a : Distance from the pivot point to the Center of Mass (G).
- I : Moment of Inertia *about the axis of rotation*, not necessarily the center of mass.

II. Derivation Methods (Textbook Alignment)

Your students must be comfortable deriving the equation of motion using Conservation of Mechanical Energy (ME).

1. The Energy Derivative Method (Preferred)

Since the system is conservative (frictionless):

$$ME = KE + PE = \text{constant}$$

$$\frac{d(ME)}{dt} = 0$$

Example: Compound Pendulum

- Kinetic Energy: $KE = \frac{1}{2}I\theta'^2$
- Gravitational Potential Energy: $GPE = Mga(1 - \cos \theta)$
 - Small Angle Approximation: $1 - \cos \theta \approx \frac{\theta^2}{2}$
 - $GPE \approx \frac{1}{2}Mga\theta^2$
- Mechanical Energy: $ME = \frac{1}{2}I\theta'^2 + \frac{1}{2}Mga\theta^2 = \text{const}$
- Differentiate w.r.t time:

$$\frac{d}{dt} \left(\frac{1}{2} I \dot{\theta}^2 \right) + \frac{d}{dt} \left(\frac{1}{2} M g a \theta^2 \right) = 0$$

$$I \dot{\theta} \ddot{\theta} + M g a \theta \dot{\theta} = 0$$

5. **Simplify:** Factor out $\dot{\theta}$ (since $\dot{\theta} \neq 0$ continuously):

$$\dot{\theta} (I \ddot{\theta} + M g a \theta) = 0$$

$$\ddot{\theta} + \frac{M g a}{I} \theta = 0$$

Example: Torsion Pendulum

1. Kinetic Energy: $KE = \frac{1}{2} I \dot{\theta}^2$
2. Elastic Potential Energy: $EPE = \frac{1}{2} C \theta^2$
3. Mechanical Energy: $ME = KE + EPE = \text{const}$
4. Result: $\ddot{\theta} + \frac{C}{I} \theta = 0$

2. The Newtonian Method (Verification)

Using the rotational analog of Newton's Second Law:

$$\sum \tau_{ext} = I \ddot{\theta}$$

$$-M g a \sin \theta = I \ddot{\theta}$$

- *Small Angle Approximation* ($\sin \theta \approx \theta$):

$$-M g a \theta = I \ddot{\theta} \implies \ddot{\theta} + \frac{M g a}{I} \theta = 0$$

III. System Analysis & Troubleshooting

Based on the problems in [Ch4 Book Probs.pdf](#), students often struggle with geometry and inertia calculations.

A. Finding 'G' (Center of Mass)

For complex compound pendulums (e.g., Problem 12: Rod with two masses), finding the distance a is the first step.

$$a = OG = \frac{\sum m_i x_i}{\sum m_i}$$

- **Tip:** Set the pivot O as the origin ($x = 0$). Masses on opposite sides have opposite signs (e.g., $+L/2$ and $-L/4$).

B. Calculating Inertia (I)

- **Additivity:** $I_{total} = I_{rod} + I_{mass1} + I_{mass2}$
- **Parallel Axis Theorem:** If I_G is given about the center of mass, and the pivot is distance d away:

$$I_{pivot} = I_G + M d^2$$

IV. The Physics of "Approximation"

For your advanced students, explicitly define the limits of the model.

1. Why $\theta \leq 10^\circ$? The model relies on Linearization.
- $\sin \theta = \theta - \frac{\theta^3}{3!} + \dots$

If $\theta = 10^\circ \approx 0.17$ rad:

$\sin(0.17) = 0.1691$

Error is $< 1\%$.

If $\theta = 30^\circ$, the restoring force is non-linear, the period increases, and the motion is Anharmonic.

2. Phase Space Trajectory

Small Angle: Plotting θ' vs θ yields a perfect Ellipse.

Large Angle: The ellipse distorts into an "eye" shape.
- V. Problem Bank Checklist
1. Essential Formulas
- | Type | Equation of Motion | Period (T_0) |
|----------|--------------------------------------|----------------------------|
| Simple | $\theta'' + \frac{g}{L}\theta = 0$ | $2\pi\sqrt{\frac{L}{g}}$ |
| Torsion | $\theta'' + \frac{C}{I}\theta = 0$ | $2\pi\sqrt{\frac{I}{C}}$ |
| Compound | $\theta'' + \frac{Mga}{I}\theta = 0$ | $2\pi\sqrt{\frac{I}{Mga}}$ |
2. Energy Conservation Checklist
- When using $ME = \text{const}$ to find velocity or position:
1. Zero Level: Define where $GPE = 0$ (usually the lowest point) or where $EPE = 0$ (equilibrium).

2. States: Equate Energy at Max displacement ($\omega = 0$) to Energy at Equilibrium ($\theta = 0$).

Max Displacement: $ME_{max} = EPE_{max}$ (Torsion) or GPE_{max} (Compound).

$EPE_{max} = \frac{1}{2}C\theta_m^2$

$GPE_{max} = \frac{1}{2}Mga\theta_m^2$

Equilibrium: $ME_{eq} = KE_{max} = \frac{1}{2}I\omega_{max}^2$

3. Result: V_{max} or ω_{max} can be found without time equations.

3. Common Pitfalls

The "a" distance: In compound pendulums, a is the distance from the pivot to the center of mass, NOT the length of the rod.

Inertia (I): Do not blindly use $\frac{1}{12}ML^2$ or $\frac{1}{3}ML^2$. Check where the axis of rotation is. Use Parallel Axis Theorem if necessary.

Units: C is in $m \cdot N$. θ is in Radians. M is in kg.